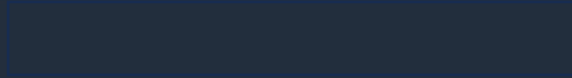


A Predictive Analysis of Amazon Best Sellers



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DATASET & RESEARCH QUESTION

What product characteristics predict Best Seller status on Amazon?

- **Target:** `isBestSeller` (binary) — only **0.41%** of products qualify
- **Continuous predictors:** price, star rating, review count — all right-skewed
- **Categorical predictor:** `category_name` — 250+ levels, high dimensionality
- **Core challenges:** extreme class imbalance; non-linear social-proof effects; sparse category structure

Pre-processing pipeline

- Filter `price > 0 & stars > 0` — removes 93.7% of raw rows
- Log-transform: `log_price = log(price)`, `log_reviews = log1p(reviews)`
- Stratified 70/30 train/test split, `set.seed(1)`

1.4M

RAW OBS.

89,957

AFTER FILTER

0.41%

BS RATE

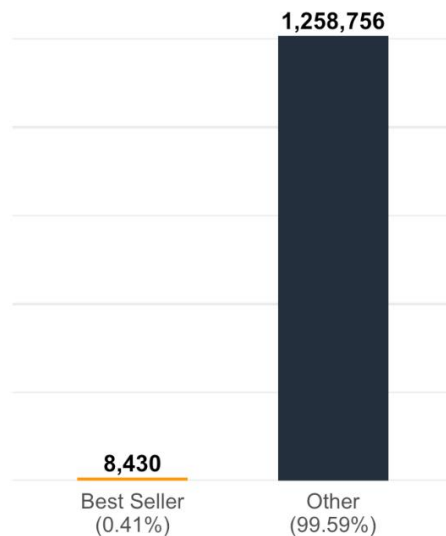
250+

CATEGORIES

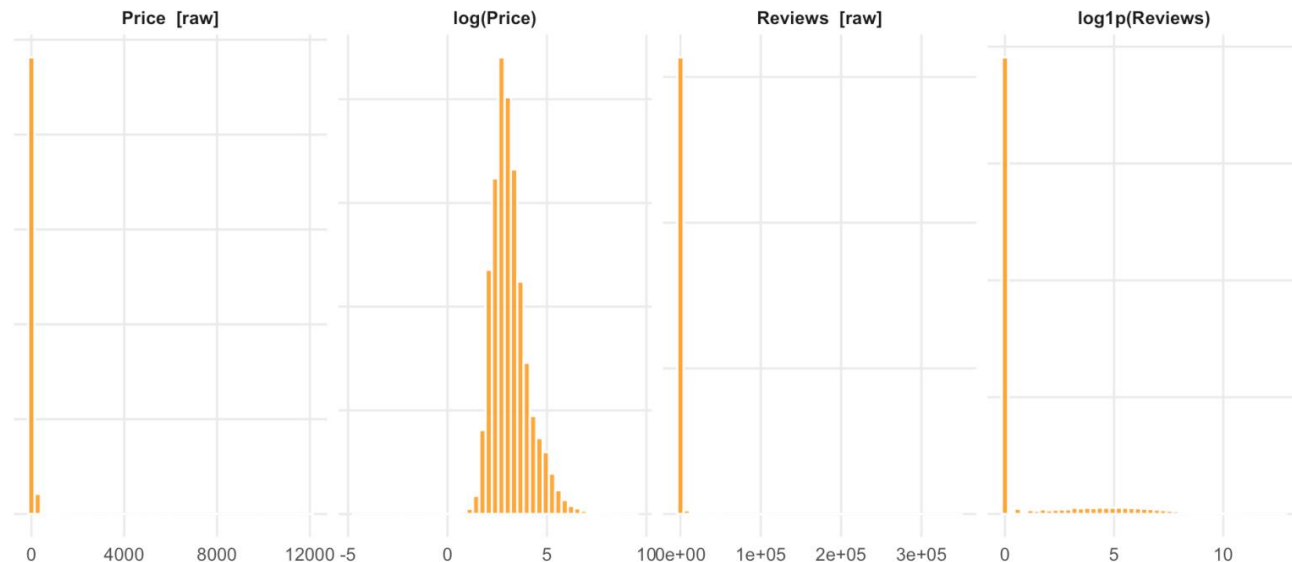
Variable	Median	Mean
Price (\$)	26.99	48.90
Stars	4.5	4.39
Reviews	3	501

EDA — CLASS IMBALANCE & PREDICTOR SKEW

Class Imbalance

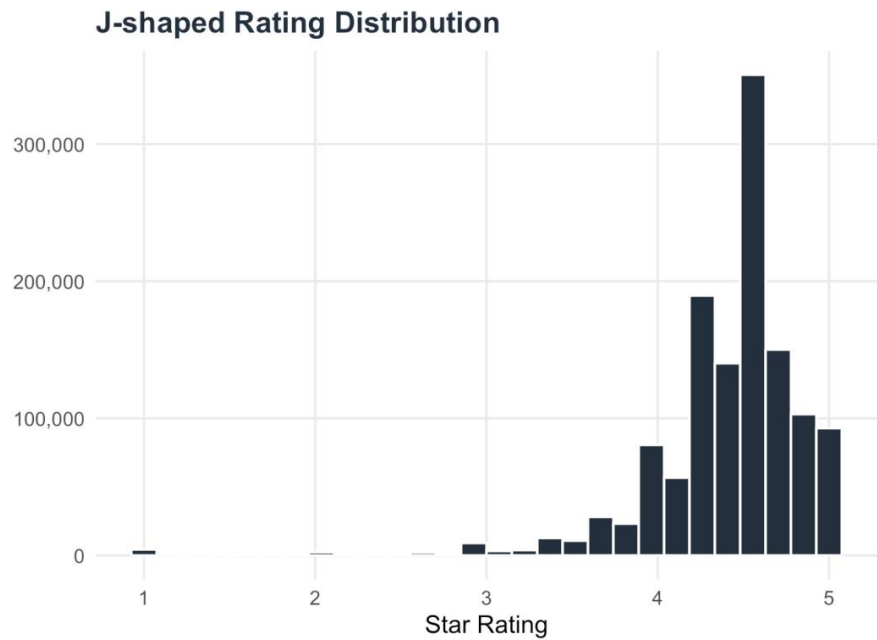


Before and After Log Transform



Both raw predictors are severely right-skewed. Log transforms are necessary for well-behaved model estimation.

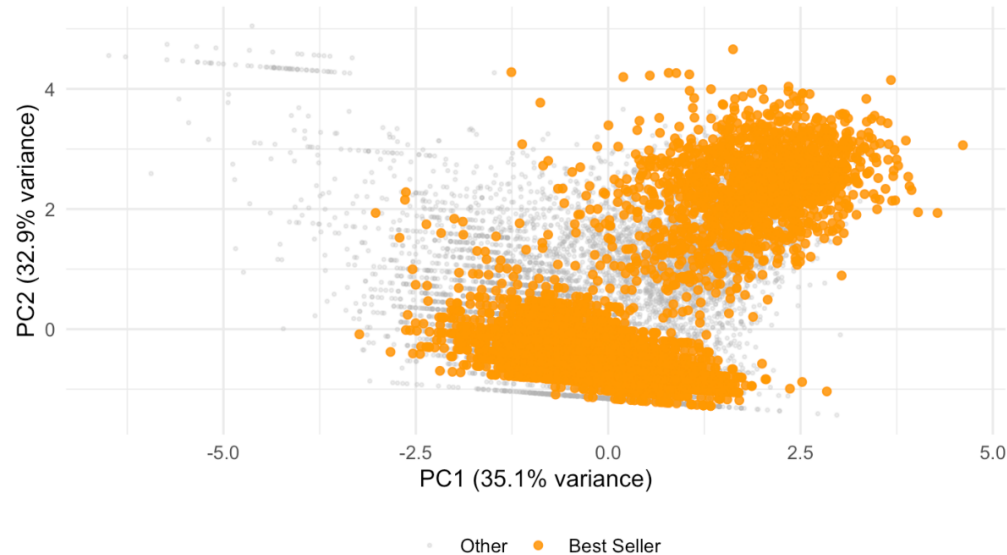
EDA — RATING DISTRIBUTION & CATEGORY PATTERNS



Star ratings cluster near 4.5 — a J-shape incompatible with Gaussian assumptions. Best Seller density varies >10× across categories.

UNSUPERVISED EXPLORATION

PCA on numeric predictors



Best Sellers **overlap** the negative-class cloud — not linearly separable in numeric space alone.

Category clustering (Ward, k = 4)

- Apparel → **lowest** BS density
- Specialty electronics → **highest** BS density
- Review volume drives cluster separation

◆ Model 1: Logistic Lasso

- L_1 penalty → **auto-selects** relevant categories from 250+
- 10-fold CV, AUC criterion
- Inverse-prevalence class weights

$$\hat{\beta} = \operatorname{argmin} \{ -(1/n) \sum [y_i x_i^T \beta - \log(1 + e^{x_i^T \beta})] + \lambda \|\beta\|_1 \}$$

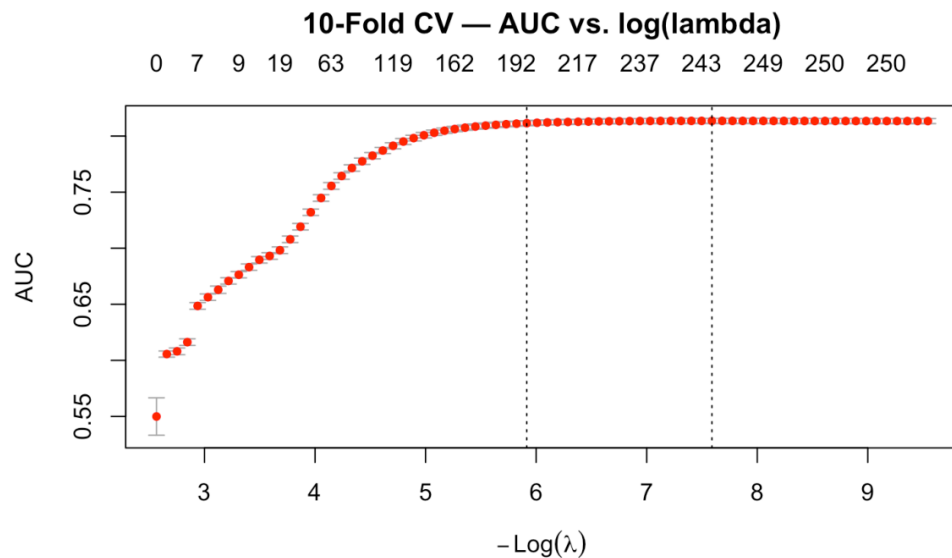
◆ Model 2: GAM + Smoothing Splines

- Captures **diminishing returns** of review volume
- No pre-specified functional form needed
- REML smoothing parameter selection

$$\operatorname{logit} P(Y=1) = \beta_0 + f_1(\log_reviews) + f_2(\log_price) + \beta_1 \cdot stars$$

Evaluated on the **30% held-out test set** — AUC and PR-AUC only (accuracy is degenerate at 0.41% prevalence).

MODEL 1 — LASSO RESULTS



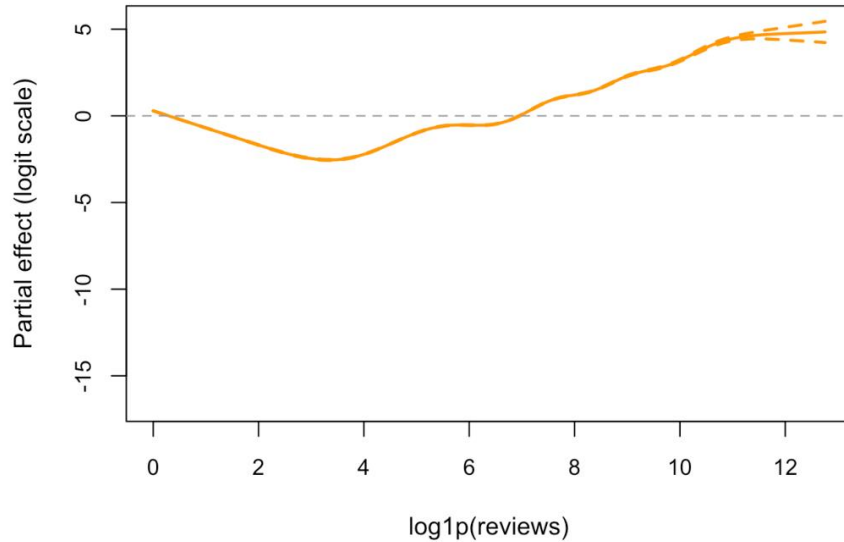
Top 8 non-zero coefficients

Term	Coeff.
cat: Nintendo Switch Consoles, Games & Accessories	-4.441
cat: Building Toys	-3.927
cat: Online Video Game Services	+3.805
cat: Kids' Play Cars & Race Cars	-3.796
cat: Laptop Bags	-3.756
cat: Headphones & Earbuds	-3.729
cat: GPS & Navigation	-3.715
cat: Tools & Home Improvement	+3.704

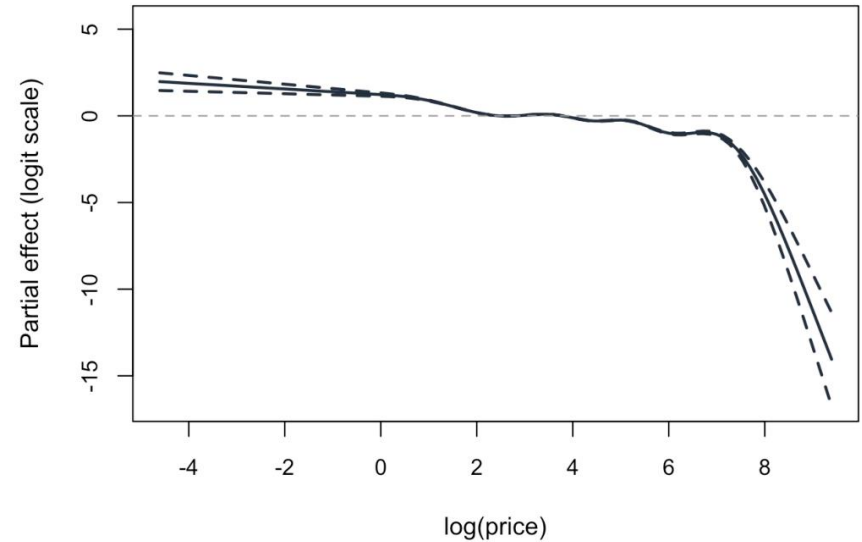
log_reviews = largest positive coefficient. Most categories zeroed out → **sparsity confirmed**.

MODEL 2 — GAM RESULTS

$f_1(\log_reviews)$ — Diminishing Returns



$f_2(\log_price)$ — Price Effect



f_1 : steep rise $\rightarrow$ plateau = diminishing returns of reviews. f_2 : lower price $\rightarrow$ higher Best Seller probability.

MODEL COMPARISON & EVALUATION

Model	ROC-AUC	PR-AUC
Lasso	0.8134	0.0435
GAM	0.6682	0.0258
Baseline (all-zero)	0.5000	0.0067

Lasso

PRIMARY MODEL (PR-AUC)

Why these metrics?

- **Accuracy excluded** — predicting all-zero = 99.59% accuracy, zero recall
- **PR-AUC** (primary) — focuses on the positive class; baseline = prevalence (0.0067)
- **ROC-AUC** (secondary) — measures ranking across all thresholds

Both models achieve a **6× lift** over the random baseline.

LIMITATIONS & DISCUSSION

Selection bias

- Filter removes 93.7% of catalog
- Excludes free listings and unrated arrivals
- Results apply to *established, priced* products only

Class imbalance

- 0.41% prevalence → inverse-prevalence weighting required
- GAM splines extrapolate at the high-review tail (sparse positives)

Apparel dominance

- 3 largest categories = apparel, lowest BS density
- Lasso coefficients estimated relative to this apparel baseline

Cross-sectional data

- Single snapshot → cannot separate stock from *flow*
- Best Seller status partly reflects sales velocity, not just product attributes

CONCLUSIONS & FUTURE WORK

Key findings

- GAM splines reveal **diminishing returns** of review volume
- **Reputation** (reviews + stars) outweighs price as the main driver
- Lasso zeroes out most categories → signal concentrated in few levels
- Apparel dominates the sample → bounds category-level interpretation

Future Work

 **Time-series panel data** — model sales velocity (flow) rather than a cross-sectional stock snapshot

 **NLP on product titles** — sentiment scores, brand signals, and category-aware embeddings to improve discrimination in the long tail of low-review products